

The data verbs: `select` · `filter` · `mutate` · `arrange`

Tuesday, June 2, 2026

i Today: Tuesday June 2

Before class

- Perusall Assignment: [DC §10.1–10.6](#)

Plan for today

- Four new data verbs: `filter`, `arrange`, `select`, `mutate`
- Sorting verbs by *what they touch*: rows vs. columns
- Combining verbs with the pipe into real pipelines
- Formatting pipelines so they stay readable
- Paying off yesterday's IOU: dropping the NA penguins

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 3](#) · [R4DS Ch. 4](#) · [DC Ch. 10](#) · [dplyr reference](#)

Recap from yesterday

You have the **framework** and your first two data verbs:

Table 1: Where we left off.

Family	Input → Output	The verb that uses it
Reduction	variable → scalar	<code>summarise()</code>
Transformation	variable → variable	<i>...today:</i> <code>mutate()</code>

Family	Input → Output	The verb that uses it
Data verb	data frame → data frame	all of them

`summarise()` is the home of reduction functions. Today you meet `mutate()`, the home of **transformation** functions, plus three verbs that reshape *which* rows and columns you keep.

A map of the verbs

R4DS §3.1.3 organizes dplyr by **what each verb operates on**:

Table 2: The dplyr toolkit.

Touches...	Verbs	What changes
Rows	<code>filter()</code> , <code>arrange()</code>	which rows, and their order
Columns	<code>select()</code> , <code>mutate()</code>	which columns, and new ones
Groups	<code>group_by()</code> , <code>summarise()</code>	(<i>yesterday</i>)

Tip

Every one of these obeys the same three rules: **first argument is a data frame, you name columns without quotes, and the output is a brand-new data frame.**

Rows: `filter()`

`filter()`: keep the rows you want

`filter()` keeps only the rows that pass a test (*DC* §10.2, R4DS §3.2.1):

```
penguins |>
  filter(species == "Gentoo")
```

```
# A tibble: 124 x 8
  species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <fct>    <fct>         <dbl>         <dbl>         <int>         <int>
1 Gentoo  Biscoe          46.1          13.2          211          4500
2 Gentoo  Biscoe           50          16.3          230          5700
3 Gentoo  Biscoe          48.7          14.1          210          4450
```

```

4 Gentoo Biscoe      50      15.2      218      5700
5 Gentoo Biscoe     47.6      14.5      215      5400
6 Gentoo Biscoe     46.5      13.5      210      4550
7 Gentoo Biscoe     45.4      14.6      211      4800
8 Gentoo Biscoe     46.7      15.3      219      5200
9 Gentoo Biscoe     43.3      13.4      209      4400
10 Gentoo Biscoe     46.8      15.4      215      5150
# i 114 more rows
# i 2 more variables: sex <fct>, year <int>

```

The test is a **transformation** that returns TRUE/FALSE for each row; `filter()` keeps the TRUES.

The comparison operators

These are the tests you’ll use inside `filter()` (DC Table 10.9):

Table 3: Comparisons for `filter()`.

Operator	Means	Example
<code>==</code>	equal to	<code>species == "Gentoo"</code>
<code>!=</code>	not equal to	<code>sex != "male"</code>
<code>> >=</code>	greater (or equal)	<code>body_mass_g >= 5000</code>
<code>< <=</code>	less (or equal)	<code>flipper_length_mm < 190</code>
<code>%in%</code>	is one of	<code>island %in% c("Dream", "Biscoe")</code>

Warning

Use `==` to **test** equality, not `=`. A single `=` means “assign,” and dplyr will stop you: “*Did you mean ==?*” This is a classic AI-generated bug too, exactly the kind of thing Friday’s lesson said to read for.

Multiple conditions

Separate tests with a comma (or `&`) for **AND**, a vertical bar `|` for **OR** (R4DS §3.2.1):

```

# AND: Adelies that are ALSO on Dream
penguins |>
  filter(species == "Adelie", island == "Dream")

```

```
# OR: any penguin that is Adelie OR Gentoo
penguins |>
  filter(species == "Adelie" | species == "Gentoo")

# %in% is the clean way to write a long OR
penguins |>
  filter(species %in% c("Adelie", "Gentoo"))
```

i Note

Watch the English-trap: `filter(species == "Adelie" | "Gentoo")` does **not** work the way it reads. Each side of `|` must be a complete test.

Paying off yesterday's IOU

Yesterday, `group_by(species, sex)` gave us an NA group because some penguins have a missing `sex`. Now we can drop them:

```
penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(n = n(), .groups = "drop")
```

```
# A tibble: 6 x 3
  species    sex      n
  <fct>    <fct> <int>
1 Adelie  female   73
2 Adelie  male     73
3 Chinstrap female   34
4 Chinstrap male     34
5 Gentoo  female   58
6 Gentoo  male     61
```

`is.na(sex)` is TRUE for the missing ones; `!` flips it, so `!is.na(sex)` keeps the rows that **do** have a sex. No more NA group.

Rows: `arrange()`

`arrange()`: set the order

`arrange()` reorders rows without adding or removing any (*DC* §10.4, *R4DS* §3.2.3). Ascending by default:

```
penguins |>
  arrange(body_mass_g)
```

```
# A tibble: 344 x 8
  species    island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <fct>      <fct>         <dbl>         <dbl>           <int>         <int>
1 Chinstrap Dream          46.9           16.6             192          2700
2 Adelie     Biscoe          36.5           16.6             181          2850
3 Adelie     Biscoe          36.4           17.1             184          2850
4 Adelie     Biscoe          34.5           18.1             187          2900
5 Adelie     Dream          33.1           16.1             178          2900
6 Adelie     Torgers~        38.6            17             188          2900
7 Chinstrap Dream          43.2           16.6             187          2900
8 Adelie     Biscoe          37.9           18.6             193          2925
9 Adelie     Dream          37.5           18.9             179          2975
10 Adelie    Dream          37            16.9             185          3000
# i 334 more rows
# i 2 more variables: sex <fct>, year <int>
```

Wrap a column in `desc()` for descending order, and list several columns to break ties:

```
penguins |>
  arrange(species, desc(body_mass_g)) # by species, then heaviest first
```

Columns: `select()`

`select()`: keep the columns you want

`select()` is to columns what `filter()` is to rows (*DC* §10.1, *R4DS* §3.3.2):

```
penguins |>
  select(species, island, body_mass_g)
```

```
# A tibble: 344 x 3
  species island   body_mass_g
  <fct>   <fct>         <int>
1 Adelie  Torgersen       3750
2 Adelie  Torgersen       3800
3 Adelie  Torgersen       3250
4 Adelie  Torgersen        NA
5 Adelie  Torgersen       3450
6 Adelie  Torgersen       3650
7 Adelie  Torgersen       3625
8 Adelie  Torgersen       4675
9 Adelie  Torgersen       3475
10 Adelie Torgersen       4250
# i 334 more rows
```

You can grab a range, drop with `!`, or rename as you go:

```
penguins |> select(species:bill_depth_mm) # a range of columns
penguins |> select(!year)                 # everything EXCEPT year
penguins |> select(mass = body_mass_g)    # rename while selecting
```

i Note

`penguins` has only 8 columns, so `select()` feels optional here. On a 19-column table like `flights` from `HW2`, or a 76-column one, it's the first thing you reach for.

select() helpers

For wide tables, name columns by pattern instead of one-by-one (R4DS §3.3.2):

```
penguins |> select(starts_with("bill")) # bill_length_mm, bill_depth_mm
penguins |> select(ends_with("_mm"))    # all the millimetre measurements
penguins |> select(where(is.numeric))   # every numeric column
```

Columns: mutate()

mutate(): add a new column

`mutate()` adds columns computed from existing ones, keeping every row (*DC* §10.3, R4DS §3.3.1):

```
penguins |>
  mutate(body_mass_kg = body_mass_g / 1000) |>
  select(species, body_mass_g, body_mass_kg)
```

```
# A tibble: 344 x 3
  species body_mass_g body_mass_kg
  <fct>      <int>      <dbl>
1 Adelie     3750         3.75
2 Adelie     3800         3.8
3 Adelie     3250         3.25
4 Adelie      NA         NA
5 Adelie     3450         3.45
6 Adelie     3650         3.65
7 Adelie     3625         3.62
8 Adelie     4675         4.68
9 Adelie     3475         3.48
10 Adelie    4250         4.25
# i 334 more rows
```

mutate() is where transformations live

This is the parallel *DC* §10.3 draws to yesterday:

`mutate()` **always** involves a *transformation* function, just as `summarise()` **always** involves a *reduction* function.

```
# transformation: one value per row → mutate()
penguins |> mutate(bill_ratio = bill_length_mm / bill_depth_mm)

# reduction: one value total → summarise()
penguins |> summarise(avg_bill = mean(bill_length_mm, na.rm = TRUE))
```

Tip

Deciding between them is the same question from yesterday: does the calculation give **one value per row** (`mutate`) or **one value for the whole group** (`summarise`)?

A categorical column with ifelse()

`ifelse()` (from *DC* §7.4) is a transformation that picks between two values per row, perfect inside `mutate()`:

```
penguins |>
  mutate(size = ifelse(body_mass_g > 4000, "large", "small")) |>
  select(species, body_mass_g, size)
```

```
# A tibble: 344 x 3
  species body_mass_g size
  <fct>      <int> <chr>
1 Adelie     3750 small
2 Adelie     3800 small
3 Adelie     3250 small
4 Adelie      NA <NA>
5 Adelie     3450 small
6 Adelie     3650 small
7 Adelie     3625 small
8 Adelie     4675 large
9 Adelie     3475 small
10 Adelie    4250 large
# i 334 more rows
```

Combining verbs

The pipe does the real work

One verb does one thing. Real questions need several, chained with `|>` (R4DS §3.4):

```
penguins |>
  filter(!is.na(sex)) |>
  mutate(body_mass_kg = body_mass_g / 1000) |>
  select(species, sex, body_mass_kg) |>
  arrange(desc(body_mass_kg))
```

Read it top to bottom as English: take penguins, **then** drop missing sex, **then** add mass in kg, **then** keep three columns, **then** sort heaviest first.

💡 Tip

Because each verb returns a new data frame, the output of one is the input to the next. The verbs line up on the left so you can skim the *story* without reading the details.

Putting all six together

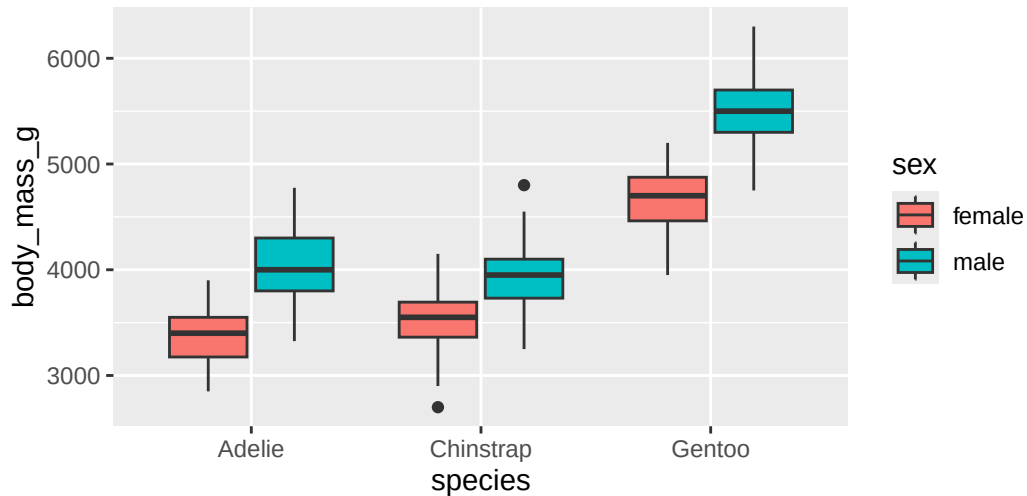
Yesterday's verbs and today's verbs, in one pipeline that answers a real question, "what's the average mass of each species/sex, biggest first?":

```
penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(
    n = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(avg_mass))
```

```
# A tibble: 6 x 4
  species sex      n avg_mass
  <fct>   <fct> <int>   <dbl>
1 Gentoo male     61  5485.
2 Gentoo female   58  4680.
3 Adelie male     73  4043.
4 Chinstrap male    34  3939.
5 Chinstrap female   34  3527.
6 Adelie  female   73  3369.
```

And because the result is just a data frame, pipe it straight into a plot (switching `|>` to `+`):

```
penguins |>
  filter(!is.na(sex)) |>
  ggplot(aes(x = species, y = body_mass_g, fill = sex)) +
  geom_boxplot()
```



Formatting your pipelines

A multi-step pipeline is only readable if you format it. The rules that matter most (R4DS Ch. 4):

- A space before `|>`, and let `|>` be the **last thing on the line**.
- For verbs with **named arguments** (`mutate`, `summarise`), put each argument on its **own line**, indented two spaces.
- Verbs without named arguments (`filter`, `select`, `arrange`) can stay on one line if they fit.
- Name new columns in `snake_case`: `body_mass_kg`, not `BodyMassKG`.

```
# Strive for                                # Avoid
penguins |>
  filter(!is.na(sex)) |>
  group_by(species) |>
  summarise(
    n    = n(),
    avg = mean(body_mass_g, na.rm = TRUE)
  )

penguins|>filter(!is.na(sex))|>group_by(
  species)|>summarize(n=n(),avg=mean(
  body_mass_g,na.rm=TRUE))
```

💡 Tip

Don't hand-format forever: the **styler** package reformats a whole file for you. In RStudio, `Cmd/Ctrl + Shift + P` → type "styler". Full guide: R4DS Ch. 4.

A shortcut you'll use constantly

Counting rows per group is so common it has its own verb, `count()` (R4DS §3.2.4):

```
penguins |> count(species)

# is exactly the same as
penguins |>
  group_by(species) |>
  summarise(n = n())
```

i Note

`count()` is just `group_by()` + `summarise(n = n())` bundled up. Add `sort = TRUE` to get the biggest groups first.

What we covered today

- **filter()** — keep rows that pass a test; `==` not `=`; combine with `,/|/%in%; !is.na()` to drop missing
- **arrange()** — reorder rows; `desc()` for descending; extra columns break ties
- **select()** — keep/drop/rename columns; `starts_with()` & friends for wide tables
- **mutate()** — add columns from transformations; the per-row twin of `summarise()`
- **The pipe** — chain verbs into a readable top-to-bottom story; pipe results into `ggplot`
- **Style** — space before `|>`, named args on their own lines, `snake_case`, let `styler` do the rest

Activity

Activity: wrangling the penguins

Roughly 25 minutes. Individually or with a neighbor.

- [Activity Canvas Link](#)

Format: a handout that walks each verb, then asks you to chain them. You'll match tasks to verbs, write `filter/arrange/select/mutate` pipelines on `penguins`, debug a `=` vs `==` error, build one multi-verb pipeline that answers a question, and restyle a deliberately unreadable pipeline.

To turn in: save your `.qmd` and rendered HTML.

Wrap-up & looking ahead

Tomorrow (Wednesday): no new reading, we put all six verbs together on a messier dataset and turn raw rows into summary tables and plots.

Upcoming Deadlines

- Thu 6/4, before class: [DC §12.1](#)
- Thu 6/4, 11:59 p.m.: [Project: Team formation](#)

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · [R4DS Ch. 3](#) · [R4DS Ch. 4](#) · [DC Ch. 10](#) · [dplyr reference](#)